BrandGuard

Brand governance before publish

Every piece of marketing content made by channel partners, agencies and freelancers is checked against the company's approved facts and brand rules before it goes live. Problems are explained in plain words, uncertain cases go to a person, and nothing is approved on a silent guess.

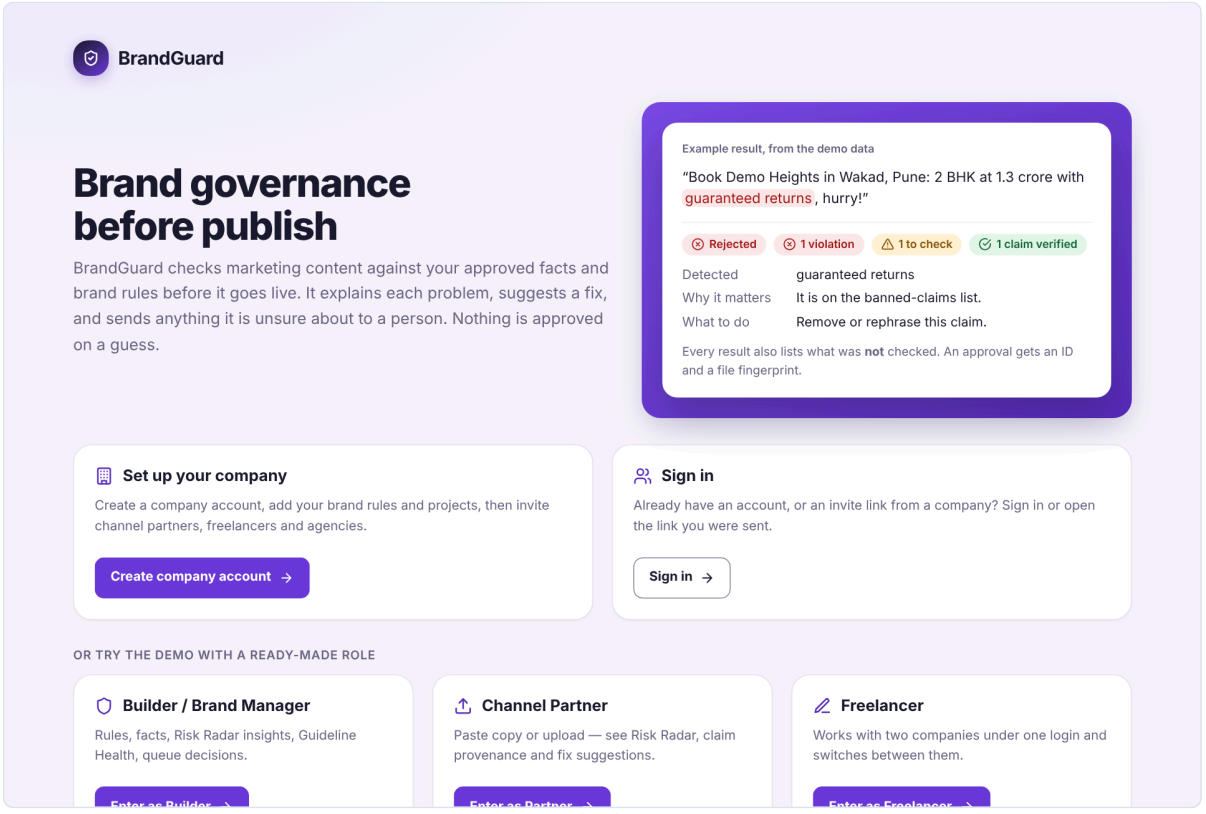
Event	CodeAxis — Round 3 product submission
Team	Sentinel-X — Anurag Gavit
Live product	**https://brandguard-55p0.onrender.com**
Repository	https://github.com/anuraggavit24/brandguard
Date	30 October 2026

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1. Try it in three minutes

No sign-up is needed. The home page has three demo roles — fictional sample accounts, kept separate from real companies.

Journey	Steps	What you should see
A creator's content is checked	Enter as **Channel Partner** → **Submit Content** → **Paste copy** → paste `Book Demo Heights in Wakad, Pune: 2 BHK at 1.3 crore with guaranteed returns, hurry!` → submit	**Rejected.** Each finding says where, why and what to change; the price is verified against the approved fact; a Risk Radar summarises the problems.
A video in Marathi	Home page → **Demo Files** → download a clip → as Channel Partner upload it under **Submit Content**	Speech is transcribed, text on screen read, every price and fact checked. A short clip takes 1–3 minutes.
The company side	Enter as **Builder** → **Review Queue** → open an item → approve or reject with a note → **Compliance Insights**	The decision is recorded in the audit trail; insights show repeated mistakes and rules that need clearer wording.
A new company, any industry	**Create company account** → choose an industry → follow the guided tour and the "Get started" checklist	Starter brand rules for the industry, sample data or spreadsheet import, invites with per-product access.



Home page: create a company, sign in, or try a ready-made demo role.

2. The problem and the solution

The problem

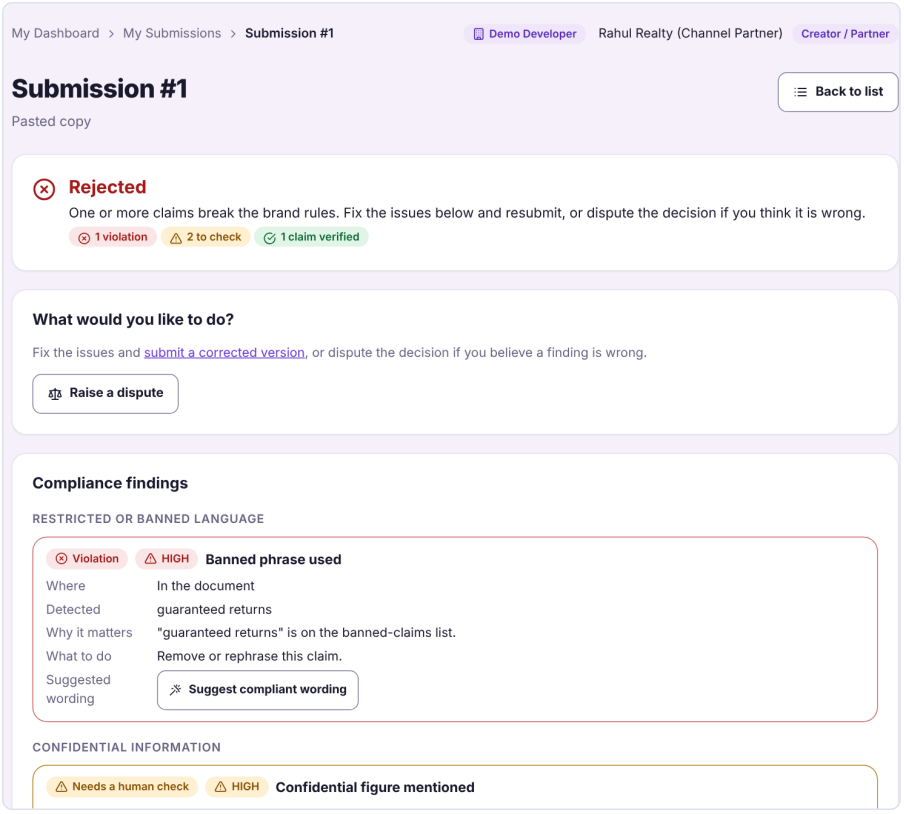
A real-estate developer (or any brand) has dozens of channel partners, agencies and freelancers producing reels, posters, brochures, captions and blog posts every day — often in Marathi and Hindi as well as English. A single wrong price, a "guaranteed returns" promise, a leaked dealer commission or an off-brand tone can create legal exposure (RERA, advertising standards) and erode trust. Today this is checked by a person watching every video, or not checked at all.

The solution

BrandGuard sits between the creator and publishing. The creator submits the content; BrandGuard reads it (speech, text on screen, document or page text), identifies the project or product, checks every claim against the company's approved, dated facts and brand rules, and returns one of five statuses:

Status	Meaning
AUTO_APPROVED	The project was identified, at least one claim was verified, and no issue is open. Gets an approval ID and a SHA-256 file fingerprint.
REJECTED	A rule is broken (wrong price, banned phrase). Every finding says where, what, why and what to change.
BLOCKED	Content for a project the creator is not allowed to promote.
SUPERVISION	BrandGuard is not certain (low recognition confidence, a reworded banned claim found by AI, a possible confidential figure). A person decides.
NEEDS_CONFIRMATION	The project could not be identified with certainty; the creator confirms which one.

Design principle. Code, approved facts and people make every decision. AI reads speech, finds reworded claims and gives advice — it never approves or rejects. Uncertainty is a status, never a pass.



A pasted caption, rejected: one violation, two items for a person to check, one price verified. Findings come first, each with where, what, why and what to do.

3. The three Round 2 improvements — promised vs delivered

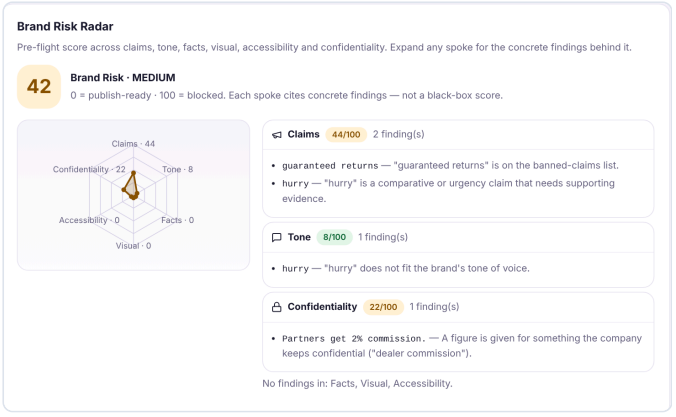
At the end of Round 2 we committed to three improvements. Each is listed with what was built, where a judge can see it, how it is tested, and what is still missing.

1 AI confidence-based matching against company rules and facts, with self-learning

Delivered	• **Recognition confidence on every finding** from speech-to-text and OCR; low-confidence words and claims go to SUPERVISION instead of passing or failing silently. • **Sound-alike matching** (`phon.py`) so misheard Marathi/Hindi words still match rules and facts (e.g. misspelled possession, bathrooms, travel time); near-miss banned phrases (edit distance) are flagged for a person. • **AI reviewer** (Groq `gpt-oss-120b`, fallback `gpt-oss-20b`): catches banned claims said in other words, tone, terminology and clarity issues, and writes a summary in the content's own language (Marathi/Hindi/English) plus English. Every quote the AI returns must exist in the transcript or it is discarded. Results are cached per transcript + rules so duplicate files cost no AI call. • **Self-learning from disputes:** when a creator disputes a finding and the builder decides, a lesson is proposed from both sides' comments. Lessons only affect wording (never prices), are inactive unless the case is clear, and the builder can switch any off. • **Suggested wording:** a rule-based suggestion when a clean one exists, or an AI rewrite on request — checked in code for banned phrases and new numbers before it is shown. • **Guideline Health:** rules that fire often or are often disputed are flagged as needing clearer wording.
Where to see it	Any submission result (findings, "Recognition confidence", AI review card); Builder → Disputes, Lessons, Compliance Insights; `/ai-status`.
Tests	`test_ai_review.py` (41), `test_learning.py` (17), `test_improvement1.py` (31), `test_spoken_variants.py` (27), `test_rewrites.py` (10), `test_llm.py` (9).
Not yet	Accuracy has not been measured on a labelled set of real clips; AI on Hindi/Marathi is weaker than on English (it only advises).



Findings: banned phrase (violation), confidential figure and an unsupported urgency claim (to a person), tone (advice).



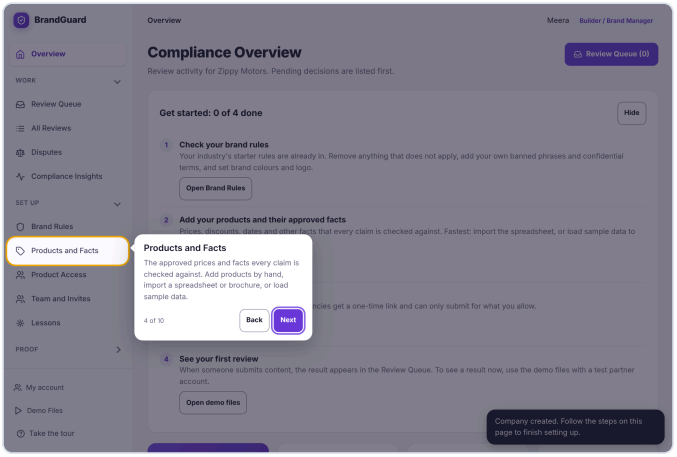
Brand Risk Radar: a 0–100 score per area, and every spoke lists the findings behind it.

2 Posters, banners, videos, blogs — and bulk upload

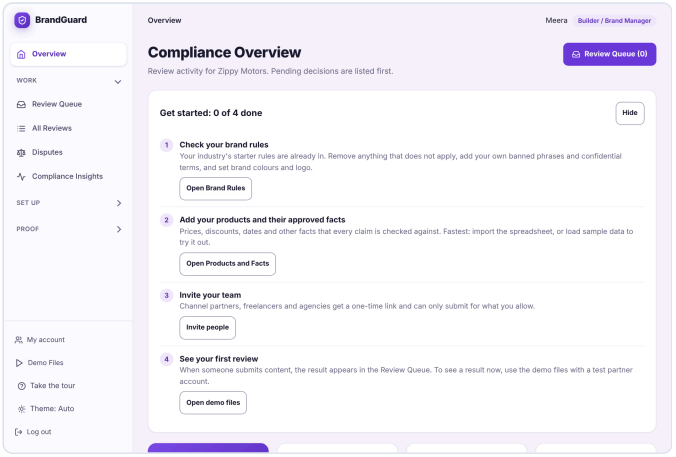
Delivered	• **Video and audio:** speech (Whisper) + text on screen (Tesseract on sampled frames and scene changes), English / Hindi / Marathi. • **Posters, banners and images:** OCR with preprocessing and four passes; a price on its own line is paired with the label above it so it is checked against the right variant. • **Documents:** PDF, Word (.docx), text. **Blogs and web pages:** fetched with SSRF protection (private addresses, redirects and size limits checked). **Pasted copy** for captions and posts. • **Brand assets and accessibility:** logo search (template match), brand colour palette, WCAG text contrast, minimum text size on images; in text: ALL CAPS, wall of text, colour-only instructions, missing alt text. • **Bulk upload:** many files at once into a background queue that survives a restart; each file gets its own result.
Where to see it	Channel Partner → Submit Content (File / Link / Paste copy tabs), Bulk Upload; Demo Files for sample clips and posters.
Tests	`test_improvement2.py` (37), `test_queue.py` (28), `test_cleanup.py` (10).
Not yet	Fonts and layout are not identified; stylised or fast-moving text on screen can be missed by OCR. YouTube/Instagram links download only from a local machine (cloud IPs are blocked by those sites).

3 Any industry — anyone creates a company and invites partners, freelancers and agencies

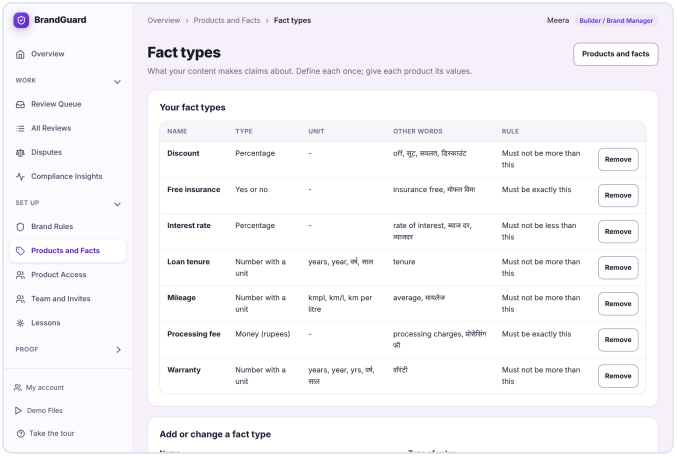
Delivered	• **Self-service company sign-up** with industry choice and **industry starter rule packs** (with sources) that produce no false alarms on clean content. • **Company-defined fact types** for any product: percent, money, number, date, yes/no, text — each with a rule (exact / must not be more / must not be less). Real estate keeps its own BHK, area, possession, RERA checks. • **Getting facts in:** a one-sheet spreadsheet template with instructions on the website, brochure import by AI (every number must appear in the document; a person confirms each row), or sample data. • **Team:** invites by role (channel partner, freelancer, agency, manager) with per-product access, "all products including later", bulk invites from a spreadsheet, emailed one-time code before adding people, email via Brevo. • **One person, many companies:** a freelancer can work for two companies under one login and switch. • **Guided tour and "Get started" checklist** for builders and creators.
Where to see it	Home → Create company account; Builder → Products and Facts, Fact types, Import, Team and Invites.
Tests	`test_improvement3.py` (56), `test_packs.py` (31), `test_accounts.py` (36), `test_onboarding.py` (13), `test_invites_bulk.py` (14), `test_safety.py` (35).
Hosting	Paid server: Render Starter (Docker) with a 2 GB persistent disk. The site does not sleep, and companies, rules, facts and reviews survive restarts and redeploys.



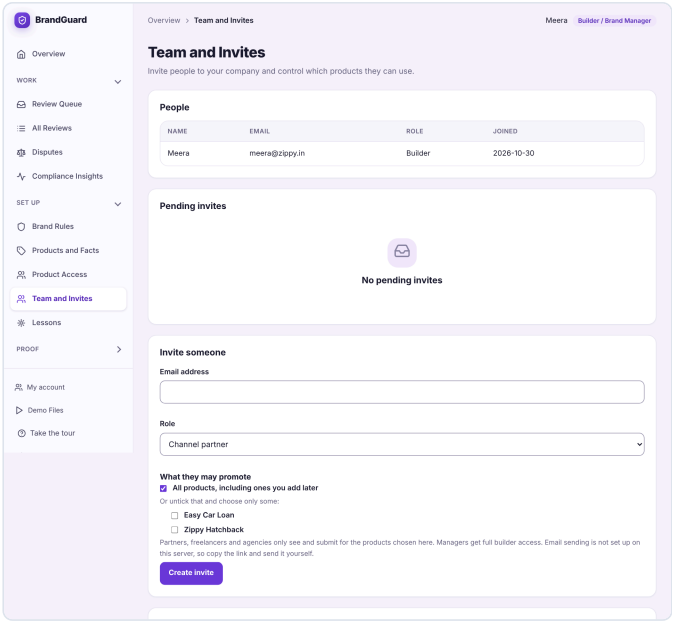
A new company (Zippy Motors, e-commerce) gets a guided tour that points at each part of the menu.



"Get started" checklist on the overview.



Company-defined fact types for a non-real-estate product.



Team and Invites: per-product access, bulk invites from a spreadsheet, reset cool-down setting.

4. Other improvements after Round 2

- **Accounts and safety:** password reset by email (single use, 60 min, company-set cool-down), sessions end after a password change, one-time codes before adding people, rate limits on sign-up, reset, suggestions and uploads, daily invite limit per company.
- **Notifications:** creator and builder are emailed when a result is ready or a decision is made (each user can switch off).
- **Indian amounts:** lakh/crore numbering and Hindi/Marathi number words ("नऊ हजार चारशे सत्तर", "दीड कोटी") are read and compared.
- **Confidential figures:** "2% commission" style leaks are flagged even when the term is phrased differently.
- **Risk Radar and claim provenance** on every result; **"What was and was not checked"** card, so a skipped check is never counted as passed.
- **Redesigned interface:** findings first on every result, grouped builder menu (Work, Set up, Proof), an example result on the home page, light and dark themes, and phone layouts without sideways scrolling.
- **Robustness:** every page and form crawled as every user with bad input; readable error pages; uploads deleted after review.
- **AI resilience:** one AI client with model fallback after Groq retired the model used in Round 2; `/ai-status` and `/ocr-status` pages for builders.

5. Complete feature list

For creators (channel partners, freelancers, agencies)

Feature	What it does
Submit content	Upload a video, audio, image, PDF, Word or text file; give a web page link; or paste copy. Only projects/products the creator may promote are offered.
Bulk upload	Many files at once; background queue with progress; each gets its own result.
Result page	Status, Risk Radar, claim-by-claim checks, claim provenance, findings with where/what/why/what to do, what was and was not checked, transcript or document text, AI summary, history.
Suggested wording	Rule-based suggestion where a clean one exists; "Ask the AI for another wording" (checked before shown); "Use this" records it as the chosen fix.
Disputes	Dispute any finding with a reason; see the builder's decision.
Verify a file	Check whether a file is exactly the approved version (SHA-256 fingerprint + approval ID).
Help, tour, demo files	Guidelines page, guided tour, sample clips and posters.
Multiple companies	One login, switch between companies the person works for.
Account	Change password, forgot password, email notification setting.

For builders and brand managers

Feature	What it does
Overview and Review Queue	Pending decisions first; approve or reject with a note; dismiss individual findings.
All Reviews, Approval Records, Audit Trail	Every submission, every approval with ID and file hash, every action with who and when.
Brand Rules	Banned phrases, confidential terms (redacted in output), preferred terminology, tones to avoid, industry starter packs with sources, optional visual and accessibility checks.
Projects / Products and Facts	Projects with unit types, prices, area, possession, RERA, offers, distances — or products with company-defined fact types. Facts are versioned and dated.
Import	One-sheet spreadsheet template (instructions on the page), brochure import with AI (person confirms each row), sample data.
Brand assets	Logo, brand colours; contrast and text-size checks on images.
Access and team	Per-project access, invites (single, all-products, bulk from spreadsheet), roles, one-time code before adding people, reset cool-down.
Disputes and Lessons	Decide disputes; lessons proposed from both sides' comments; switch any lesson off.
Compliance Insights	Most common findings, where issues appear, repeated wording flags, creators with most rejections, Guideline Health.
Status pages	`/ai-status`, `/ocr-status`, `/health`.

Checks performed on every piece of content

Project/product identification (name, alias, variant, RERA, locality — never guessed) · prices and amounts (Indian numbering, Hindi/Marathi words) · area, bathrooms, possession, RERA format, offers, distance and travel time · company fact types · banned phrases (exact, sound-alike, near-miss, reworded by AI) · confidential terms and figures · comparative/urgency claims needing evidence · tone and terminology · contact details · logo, colours, contrast, text size · text accessibility · lessons from earlier disputes.

6. Technical architecture and implementation

One FastAPI process with server-rendered HTML and SQLite — no front-end build step, so anyone can clone and run it. About 14,800 lines of Python across 22 modules plus 18 test files.

```
content (file / link / web page / pasted text)
  → read it:  speech-to-text (Whisper) · text on screen (Tesseract: frames + scene changes) · document / page text
  → redact confidential terms
  → identify the project or product (name, alias, variant, RERA, locality) — unclear → creator confirms
  → deterministic checks: prices, facts, own fact types, banned / confidential / tone / terminology,
                          contacts, brand assets and accessibility
  → lessons from earlier dispute decisions (wording only, never prices)
  → AI reviewer (optional): reworded banned claims → a person checks;  tone, clarity → advice only
  → status: AUTO_APPROVED · REJECTED · BLOCKED · SUPERVISION · NEEDS_CONFIRMATION
  → creator and builder emailed · person decides anything uncertain · approval ID + SHA-256 hash
```

Module	Responsibility
`main.py`	Routes, sessions, review orchestration, statuses, audit, pages
`ui.py`	Design system, app shell, guided tour, components
`engine.py`	Pure rule checks and number classification (no database)
`amounts.py` · `phon.py`	Indian numbering and Hindi/Marathi number words · sound-alike matching for speech-to-text
`realestate.py` · `generic.py`	Projects, unit types, dated facts, access, identification · company-defined fact types and matcher
`ocr.py` · `formats.py` · `assets.py`	Frames and images · documents and SSRF-safe page fetching · logo, colours, contrast, accessibility
`ai_review.py` · `rewrite.py` · `learning.py` · `llm.py`	AI reviewer (cached, verified) · suggestions · lessons from disputes · one AI client with model fallback
`governance.py`	Risk Radar, claim provenance, Guideline Health
`packs.py` · `importer.py`	Industry starter rules with sources · spreadsheet and brochure import
`accounts.py` · `mailer.py` · `ratelimit.py` · `jobs.py`	Accounts, invites, OTP, password reset · email (Brevo / SMTP) · rate limits · bulk queue

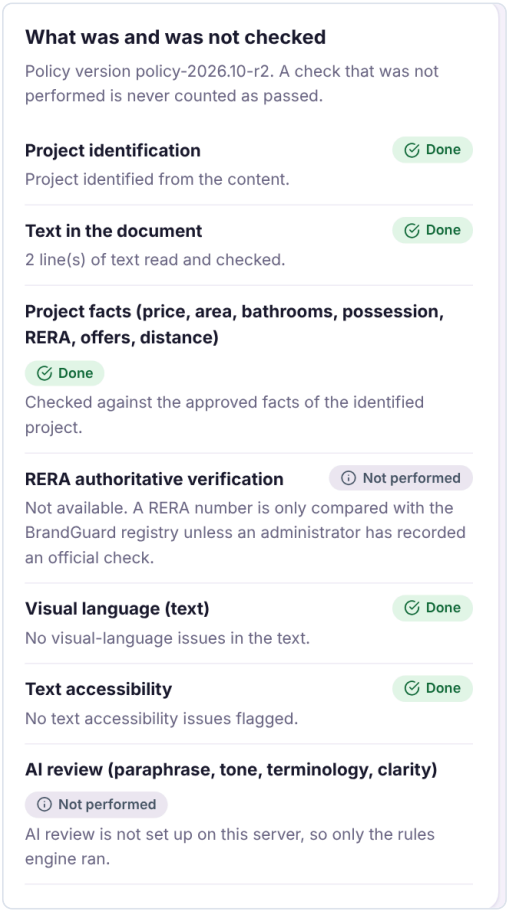
Where AI is used — and where it is not

Task	Tool	Can it decide?
Speech to text	Groq `whisper-large-v3` (or local faster-whisper)	No — numbers are re-read by code
Reworded banned claims, tone, summary	Groq `openai/gpt-oss-120b` (fallback `gpt-oss-20b`)	No — quotes must exist in the transcript; matches go to a person
Rewrite suggestions	same	No — checked for banned phrases and new numbers before shown
Lessons from disputes	same	No — inactive unless clear; builder can switch off
Brochure import	same	No — every number must appear in the document; a person confirms each row
Text on screen	Tesseract OCR + OpenCV preprocessing	No

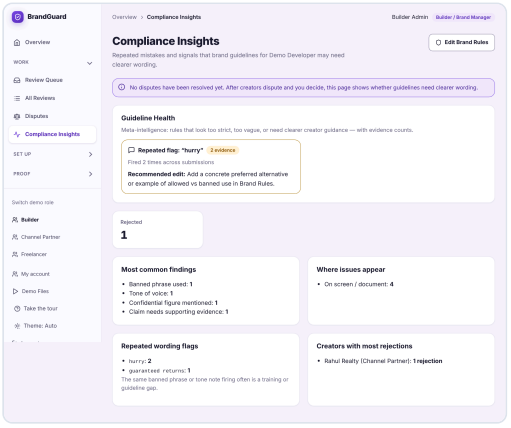
Data and deployment

- **Storage:** SQLite in `DATA_DIR`. Facts are versioned with validity dates, so an old submission still shows the fact it was checked against. Uploaded files are deleted after review unless `KEEP_UPLOADS=1`.
- **Deployment:** Docker image (Python 3.11, ffmpeg, Tesseract with English, Hindi and Marathi) on Render. Every setting has a default in `.env.example`; every key is optional — without an AI key the site, rules, facts, paste copy, documents and the full workflow still work.
- **Background work:** bulk queue in the database, resumed after a restart; emails sent on a background thread.
- **Cost control:** AI results cached per transcript + rules + prompt version; daily per-company AI limit; free-tier models.

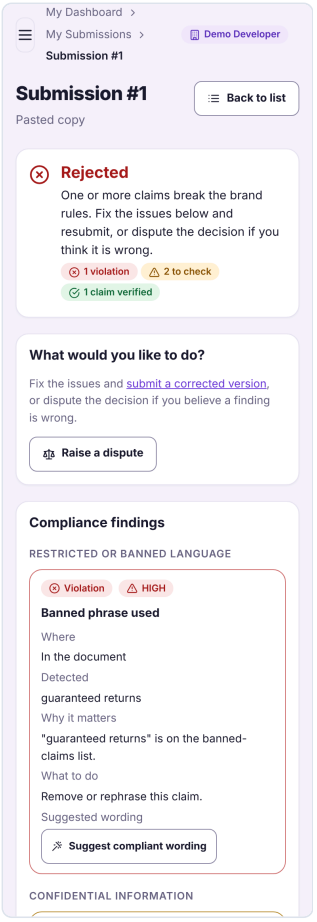
Examples from the product



A check that was not performed (RERA register lookup, AI review without a key) is shown as "Not performed" — never counted as passed.



Compliance Insights and Guideline Health for the builder.



The same result on a phone.

7. Testing details

519
automated checks, all passing on Python 3.11

18
test files + engine self-test (25)

102
routes crawled as 6 kinds of user

0
server errors on bad input after fixes

File	Checks	Covers
`test_brandguard.py`	41	Project identification, price and fact checks, access control, auto-approval rule, fact versions, migration
`test_judge_crawl.py`	18	All 102 routes as visitor, builder, partner, freelancer, new and empty company — empty / garbage / script / huge input, wrong ids, broken files, double actions, empty states: no crash, no hang, no unescaped script
`test_spoken_variants.py`	27	Real Marathi clip: misspelled possession, bathrooms, travel time, booking; misread banned phrases; commission figures
`test_amounts.py`	40	Indian numbering, Hindi/Marathi number words, percentages
`test_improvement1.py` · `test_learning.py` · `test_rewrites.py`	31 · 17 · 10	Tone and terminology, suggestions, speech clarity, replay; lessons from disputes; clean offline suggestions
`test_ai_review.py` · `test_llm.py`	41 · 9	AI output verification, caching, language, failure handling; model fallback
`test_improvement2.py`	37	Posters, documents, web pages (SSRF-safe), contrast, logo, colours, OCR pairing
`test_improvement3.py` · `test_packs.py`	56 · 31	Any-industry facts, spreadsheet and brochure import, sample data; packs without false alarms
`test_accounts.py` · `test_onboarding.py` · `test_invites_bulk.py` · `test_safety.py`	36 · 13 · 14 · 35	Sign-up, login, invites, tour and checklist, bulk invites, OTP, password reset, rate limits, notifications
`test_queue.py` · `test_cleanup.py`	28 · 10	Bulk queue and restart recovery; upload clean-up, menu labels

How we tested

- **Deterministic tests:** speech and AI are replaced by scripted answers, so results do not depend on a network or a model version. Each test file runs on its own (`python test_x.py`) and prints PASS/FAIL.
- **Robustness crawler:** found 22 server errors (huge numbers and ids, infinite prices) in our own code before submission; all fixed, and the crawler now guards against regressions.
- **Measured on real content:** 50 test items checked by hand against the expected result — **38 of 50 (76%) correct**. Six demo posters/banners and three industry spreadsheets were also run through the full pipeline and give the expected results.
- **Manual testing on the live site:** real Marathi, Hindi and English clips, posters and documents; sign-up, invites and password reset with real email.
- **Bugs found and fixed during Round 3** (examples): a retired AI model broke reviews → fallback client; Devanagari vowel signs were stripped in matching; OCR misreads approved content → low-confidence text now goes to a person; a confidential "2% commission" in a poster → dedicated figure check; Python 3.11 f-string crash on deploy → every change now verified on 3.11.

8. Reliability and security

Area	Measure
Bad input	All forms validate input; huge numbers, wrong ids and broken files give a clear message or a 404 page, never a crash; a readable error page is the last safety net.
Double actions	Deciding, disputing or accepting an invite twice is handled (second action is a no-op with a message).
Empty states	Every list page has an empty state with the next step; a new company has a checklist.
AI failure	If the AI is unreachable, the rules engine result stands and the page says the AI review was not performed. A rule-based suggestion is kept if an AI rewrite fails.
Passwords and sessions	scrypt hashing; sessions signed; all sessions end after a password change; reset links single use, 60 minutes, company cool-down.
Adding people	One-time code emailed to the builder (10 minutes, 5 tries, valid for 30 minutes of the session); a fresh code for every bulk send; 100 invites/day per company.
Abuse limits	Sign-up 5/hour/IP, password reset 10/hour/IP, AI suggestions 30/hour/user, uploads 30/hour/person.
Web fetching	SSRF protection: private and local addresses refused, redirects re-checked, size and time limits.
Output safety	All user content escaped (crawler checks for unescaped script); confidential terms redacted in output.
Secrets	API keys only in server environment variables; none in the repository.
Data	Uploaded files deleted after review; access limited per project; demo data separate from real companies.

9. Known limitations

- **Accuracy is measured only on a small set:** 38 of 50 hand-checked items (76%). A larger labelled set is the first roadmap item.
- **Speech-to-text and OCR make mistakes.** Prices are re-checked by code and low confidence goes to a person, but stylised or fast text on screen can be missed.
- **AI on Hindi and Marathi** is weaker than on English; it only advises or sends to a person.
- **Visual checks are limited** to logo, colours, contrast and text size. Fonts and layout are not identified.
- **RERA numbers are format-checked**, not looked up on the state register (no stable public API).
- **Single process and SQLite** — right for a pilot; videos are reviewed one at a time. Production at scale would use a managed database and several workers.
- **Link download** (YouTube / Instagram) works from a local machine only.

10. Roadmap

When	What	Why
Next 1 month	Labelled test set of 200 real clips and posters; publish precision/recall per check. Daily backups of the database; uptime monitoring.	Turn "works on our tests" into measured accuracy; no data loss.
1–3 months	PostgreSQL and a proper job queue (multiple workers); WhatsApp and Instagram caption checks before posting; reviewer mobile view; per-company AI budget dashboard.	Scale beyond a pilot; meet creators where they publish.
3–6 months	RERA register lookup where states offer it; font and layout checks on posters; Hindi/Marathi-tuned models for the AI reviewer; SSO for agencies.	Close the biggest known gaps.
6–12 months	Public API and webhooks (check before scheduling a post); integrations with social schedulers and DAMs; more industry packs (BFSI, pharma, auto) reviewed by domain experts.	Become the check every publishing tool calls.

Business model. Per-company subscription by number of creators and reviews per month; free tier for small builders; AI usage included up to a cap.

11. Tools and AI used

Category	Used
Language and framework	Python 3.11, FastAPI, Uvicorn, server-rendered HTML (no front-end build)
Data	SQLite; openpyxl (spreadsheets); python-docx, pypdf (documents)
Speech	Groq Whisper `whisper-large-v3` (optional local faster-whisper); ffmpeg
Vision	Tesseract OCR (eng, hin, mar), OpenCV, NumPy
Language model	Groq `openai/gpt-oss-120b` with fallback `openai/gpt-oss-20b` — advice only
Email	Brevo transactional email API (SMTP supported)
Hosting	Docker, Render Starter with persistent disk
Development	Development was assisted by Claude (Anthropic) as a coding assistant. Product decisions, testing on real content and deployment were done by the team.

Links. Live: https://brandguard-55p0.onrender.com · Repository: https://github.com/anuraggavit24/brandguard · Clone-and-run instructions, Docker and Render deployment in `README.md`; design rationale in `SOLUTION.md`.